

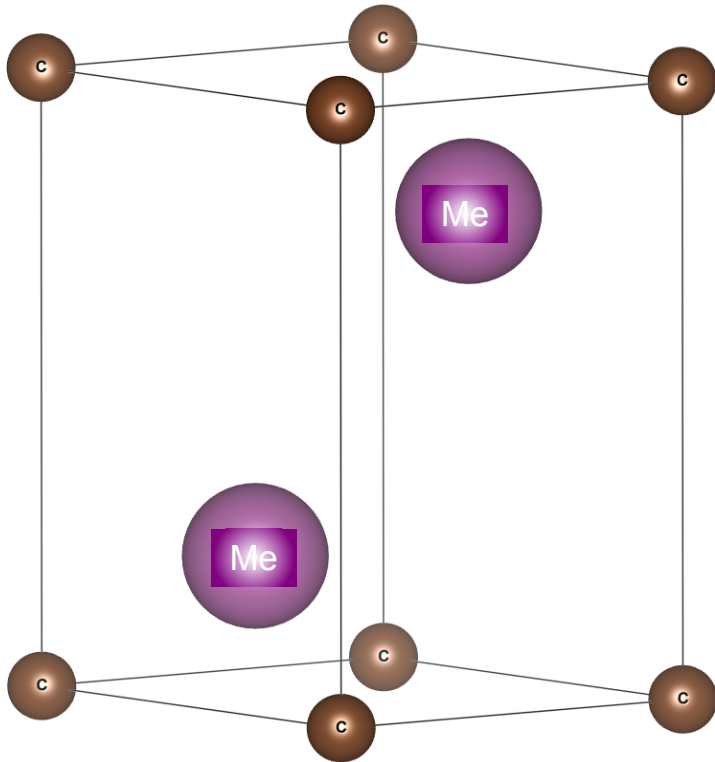
LAB Course ThC

Solid State Project:

Carbides:

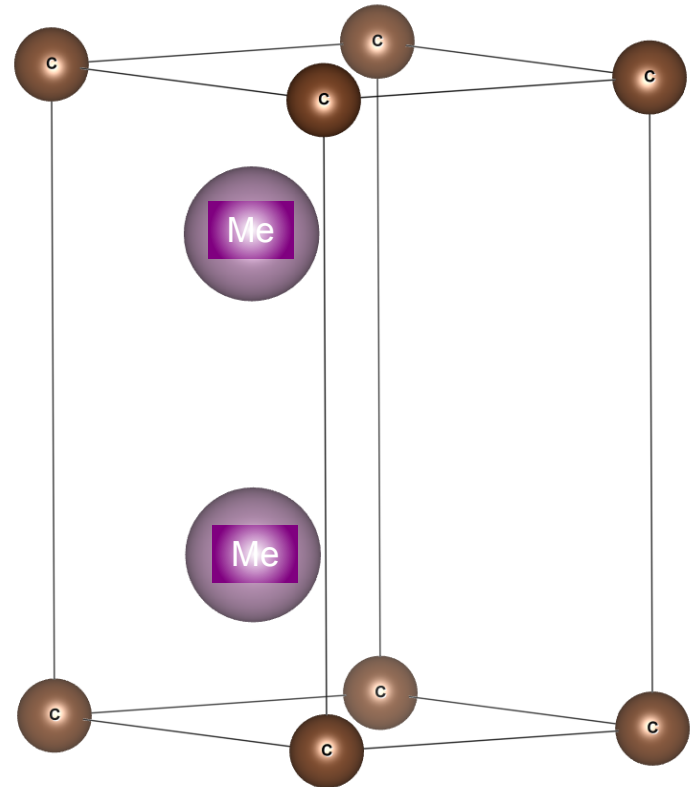
Geometry,
DOS and Band Structure

Carbides Me_2C



Me_2C (trigonal)

C: simple hexagonal
Me: approx. on cell diagonal



Me_2C (hexagonal)

C: simple hexagonal
Me: stacked

1. Preoptimization

- set up a unit cell for your system (use $a = 3 \text{ \AA}$ and $c = 6 \text{ \AA}$ as start lattice constants)
- for all following calculations use:
PREC = Accurate, and EDIFF = 1E-05
- Find out whether your material is spin-polarized in the ground state:
 - Perform a full geometry optimization (atom positions + cell) with and without spin polarization adopting
 - a reasonable cutoff energy (!)
 - $(10 \times 10 \times 5)$ **k**-mesh (Gamma)
 - Methfessel-Paxton smearing of order 1, $\sigma = 0.2$
- ***Report the pre-optimized cell parameters!***
- ***Compare total energies***
- ***What is the magnetization in case of the spin polarized calculation?***
- ***Decide whether you need spin polarization for your material! Use this setting in all remaining calculations!***

2. k-point convergence

- Perform single point calculations **for the in 1. obtained geometry** using the standard cutoff energy for Γ -including $(N \times N \times N/2)$ **k**-meshes where $N = 5, 10, 15, 20, 25$ with
 - LT-C method
 - Methfessel-Paxton smearing of order 1 with $\sigma = 1.0$ **and** 0.2
- *Tabulate and plot $E(\sigma \rightarrow 0)$ vs. N for the tested smearings (all in one graph)*
- *Compare the convergence behavior of the three k-point integration schemes!*

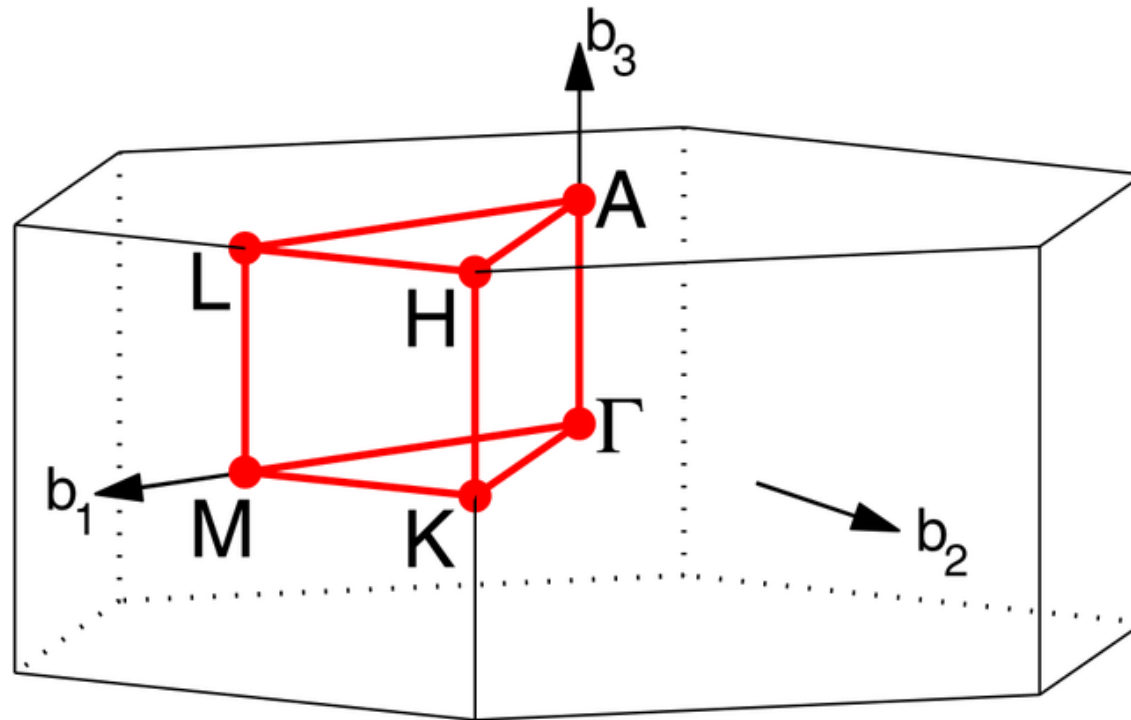
3. Geometry Relaxation

- Perform a final geometry relaxation (`EDIFFG = -0.01`) using
 - a reasonable cutoff energy
 - reasonable `ISIF`
 - Methfessel-Paxton smearing of order 1, $\sigma = 0.2$
 - a converged **k**-mesh (criterion: 1 meV per atom)
 - start from the pre-optimized geometry!
- *Report the final optimized lattice constant(s)!*
- *Report the final energy!*
- *In case of spin polarization: What is the magnetization per unit cell?*

4. DOS and Band Structure

- Calculate the DOS for the **optimized geometry of 3.**
 - keep **k**-mesh from geometry optimization for DOS
 - use LT-C smearing for DOS
 - set **NEDOS = 1000** in INCAR
 - you may reduce the energy cutoff to the standard value
- Calculate the band structure for the **optimized geometry**
 - use 20 **k**-points per line for the band structure
 - keep density from the DOS calculation fixed! (**ICHARG = 11**)
 - Band structure path (→ use xcrysden to determine coordinates):
 Γ –M–K– Γ –A–L–H–A | L–M | K–H
- ***Plot DOS and band structure for the valence states! (reasonable viewport!!)***
- ***Is the system metallic or non-metallic? Size of the band gap?***

k-path: Nomenclature



HEX path: Γ -M-K- Γ -A-L-H-A|L-M|K-H

Some final Tips...

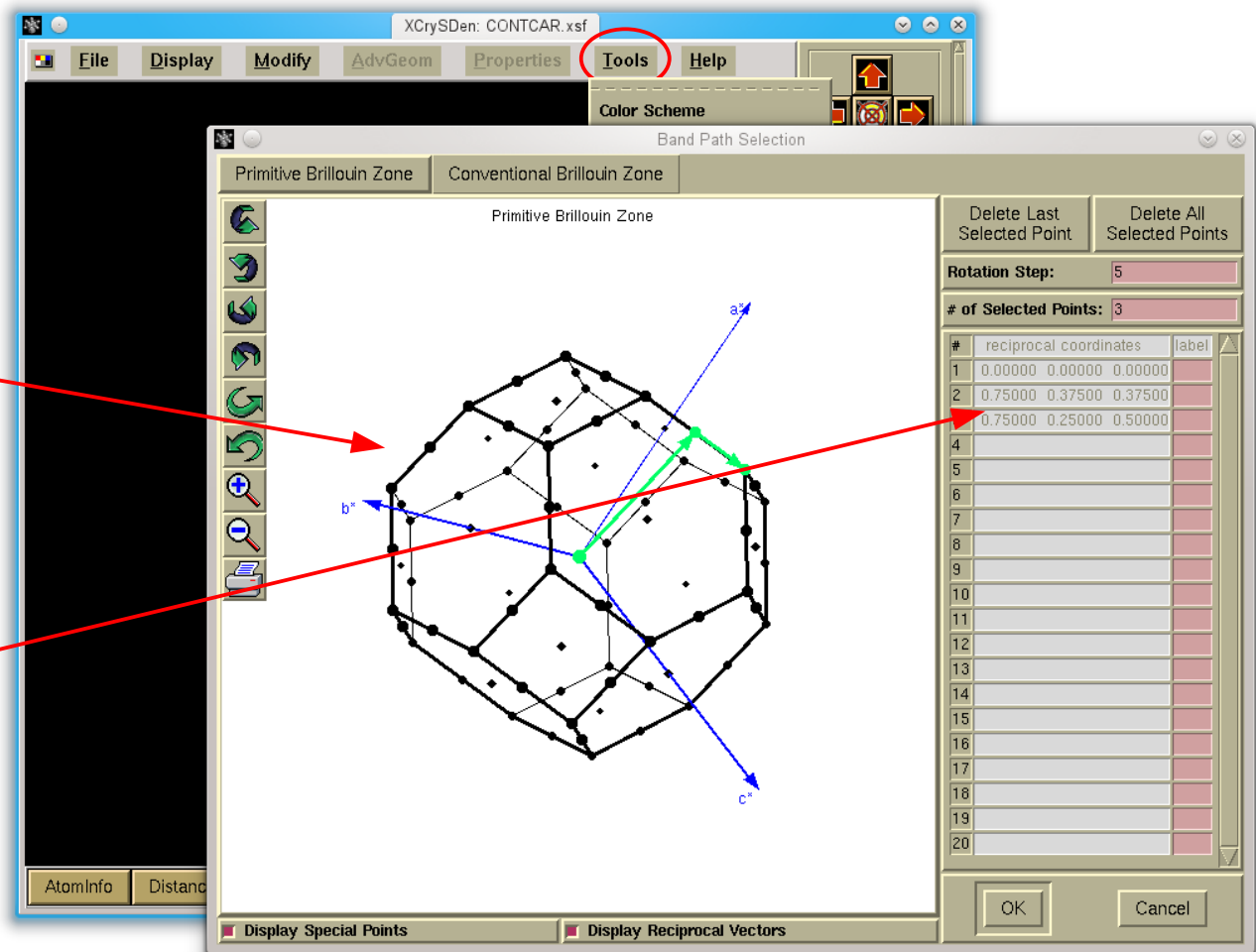
- do not run more than two calculations at one time!
- **Protocol:**
 - **submit via StudON in pdf format!**
 - **Deadline: 21 May 2026**
 - Name, student-ID and calculated system on the front page!
(but separate title page not needed)
 - short method section(s)
 - indicate all relevant information that someone else could reproduce your calculations based upon your report!
 - total energies: report 5 decimals
 - all points printed in *italic* on the preceding slides have to be addressed in the protocol!

Selecting k-paths: xcrysden

- xcrysden offers a graphical interface to determine the coordinates of special points in the BZ
- use v2xsf to convert the CONTCAR (or POSCAR) file to xsf-format (will be automatically gzipped):
`v2xsf <file>`
- open it using **xcrysden**:

click on desired
points
in the BZ

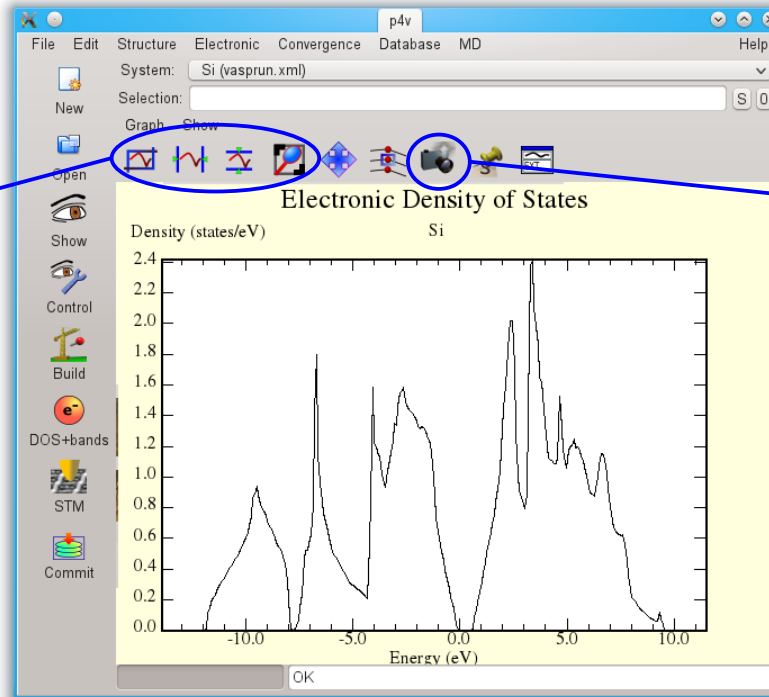
corresponding
coordinates
are displayed



p4v: Saving Graphs

- generate ps-file:

prepare
view



save current view
to
graphXXXX.ps

- convert to pdf format:
`epstopdf graph0001.ps`